FA-XFormer

Frequency-Aware Explainable Transformer for Subject-Independent EEG Emotion Recognition

Detailed Architecture and Research Workflow

Core components	Frequency-aware modeling • Transformer • Subject-independent evaluation • Neurophysiologically m
Dataset	SEED (recommended primary dataset)
Classes	Positive • Neutral • Negative
Evaluation	Leave-One-Subject-Out (LOSO) with strict leakage prevention

1. Overall Architecture

SEED EEG → Pre-processing → Temporal + Frequency + Spatial Representation → Frequency Attention → Gated Fusion → Transformer Encoder → Emotion Classification → Neurophysiological XAI → Explanation Stability and Faithfulness.

The framework should be treated as an integrated research pipeline rather than a conventional classifier with an explanation tool added after training.

2. Dataset and Subject-Independent Evaluation

Use SEED as the primary dataset. Evaluate at the subject level rather than randomly splitting EEG windows. Split subjects first, then create training/test windows. The held-out subject must not influence normalization, feature selection, hyperparameter tuning, model fitting, or explanation fitting.

3. Pre-processing Pipeline

Raw EEG → filtering → artifact handling → bad-channel handling → re-referencing → normalization → window segmentation.

Step	Starting point	Purpose
Filtering	1–45 Hz band-pass	Reduce slow drift and high-frequency noise
Artifact handling	Dataset-appropriate procedure	Reduce ocular, muscle, and other artifacts
Normalization	Training-subject statistics only	Prevent leakage
Windowing	Example: 4-second windows	Create model-ready temporal segments

4. Frequency-Aware Module

Represent EEG through frequency-specific components instead of sending the raw signal directly to the Transformer. Starting bands: Delta 1–4 Hz, Theta 4–8 Hz, Alpha 8–13 Hz, Beta 13–30 Hz, Gamma 30–45 Hz. Treat these ranges as starting points and validate them experimentally.

Preferred research direction: use learnable frequency filters while retaining conventional bands for interpretation and comparison.

EEG → Frequency decomposition → Band-specific encoders → Frequency attention → Weighted spectral representation

5. Frequency Attention

Learn an importance weight for each frequency representation so the model can identify which spectral information contributes to each prediction. Do not treat attention weights alone as definitive explanations. Validate explanations using attribution and perturbation methods.

6. Spatial / Channel Representation

EEG channels encode spatial information. Add a channel encoder or spatial attention mechanism to learn which electrodes contribute most strongly. Channel importance can later be mapped to actual scalp locations for visualization.

7. Temporal Encoder

Use a lightweight 1D CNN or TCN to capture local temporal patterns. Example: Conv1D → normalization → activation → Conv1D → temporal feature representation.

8. Multi-Domain Gated Fusion

Combine temporal, frequency, and spatial representations using a learnable gated fusion mechanism rather than simple concatenation.

Temporal features + Frequency features + Spatial features → Gated Fusion → Unified EEG representation

9. Transformer Encoder

Feed the fused representation into a compact Transformer. A practical starting point is embedding dimension 128 or 256, 4–8 attention heads, and 2–4 Transformer blocks. Tune these values only within the training data for each subject-independent protocol.

Fused representation → Positional encoding → Multi-head self-attention → Add & Norm → Feed-forward network → Add & Norm → Transformer blocks

10. Classification Head

Transformer representation → Global pooling → Dense(128) → Dropout → Dense(3) → Softmax. Outputs correspond to Positive, Neutral, and Negative emotion classes.

11. Neurophysiologically Meaningful XAI

The explanation stage should answer four questions: Which time segment mattered? Which frequency band mattered? Which EEG channels mattered? How do these factors combine for the final prediction?

Level	What to report
Temporal	Important time windows or segments
Frequency	Importance of Delta/Theta/Alpha/Beta/Gamma or learned bands
Spatial	Important channels and scalp regions
Integrated	Time × Frequency × Channel contribution

12. XAI Methods

Use complementary approaches such as Integrated Gradients, occlusion/perturbation analysis, and Transformer attention analysis. Use attention as supporting evidence rather than as the sole explanation method.

Model prediction → Attribution + Perturbation + Attention analysis → Explanation fusion → Temporal/Frequency/Channel importance

13. Scalp and Emotion-Specific Visualization

Map channel importance onto the actual scalp layout to create emotion-specific topographic visualizations. Report these as model-derived explanations and avoid unsupported causal claims about brain function.

14. Explanation Stability

Test whether explanations remain consistent across random seeds, repeated trials, similar emotional samples, subjects, and small input perturbations.

Model → Seed 1 explanation E1; Seed 2 → E2; Seed 3 → E3; Seed 4 → E4; Seed 5 → E5 → explanation agreement → stability score

Candidate similarity measures include Spearman correlation, cosine similarity, Jaccard/top-k overlap, and rank correlation. If you propose a new composite stability metric, establish its novelty through a focused literature review first.

15. Explanation Faithfulness

A stable explanation can still be wrong. Remove or mask the most important features and compare the change in prediction confidence against random-feature removal.

Condition	Expected interpretation
Original input	Baseline confidence
Remove top-ranked important features	Substantial confidence decrease supports faithfulness
Remove random features	Smaller decrease is expected

16. Complete Research Workflow

1. SEED EEG dataset
2. Subject-level train/test split

3. Pre-processing and artifact handling
4. Window segmentation
5. Temporal encoder
6. Frequency decomposition and frequency-aware encoder
7. Frequency attention
8. Spatial/channel encoder
9. Gated temporal–frequency–spatial fusion
10. Transformer encoder
11. Emotion classification
12. XAI using attribution, perturbation, and attention analysis
13. Neurophysiological visualization
14. Explanation stability analysis
15. Explanation faithfulness analysis
16. Subject-independent performance evaluation
17. Ablation studies
18. Statistical analysis and interpretation

17. Recommended Experiments

Experiment	Purpose
Baseline comparison	Compare SVM, Random Forest, CNN, LSTM, CNN-LSTM, EEGNet, Transformer, and the proposed framew
Ablation study	Remove frequency, spatial, or Transformer components to quantify their contribution.
Frequency ablation	Test the effect of removing each conventional frequency band.
Subject-independent evaluation	Use LOSO so each subject is unseen during testing.
XAI stability	Measure consistency across seeds, trials, subjects, and perturbations.
XAI faithfulness	Compare removal of important features with removal of random features.
Statistical analysis	Compare subject-level results using suitable significance tests and effect sizes.

18. Evaluation Metrics

Classification: Accuracy, Macro-F1, Precision, Recall, Cohen's kappa, balanced accuracy, confusion matrix, and where appropriate AUC. Report per-subject performance and mean ± standard deviation. Explainability: explanation stability, explanation faithfulness, top-k channel overlap, frequency-band consistency, and perturbation sensitivity.

19. Suggested Research Questions

• RQ1: Can frequency-aware representation learning improve subject-independent EEG emotion recognition compared with conventional temporal models and Transformers?

• RQ2: Can the proposed model identify consistent temporal, spectral, and spatial EEG patterns associated with different emotional states?

• RQ3: Are the generated explanations stable, faithful, and neurophysiologically meaningful across subjects and experimental conditions?

20. Proposed Contribution Positioning

A suitable positioning statement is: “We propose a frequency-aware explainable Transformer framework for subject-independent EEG emotion recognition that jointly models temporal, spectral, and spatial EEG representations and evaluates explanations through neurophysiological relevance, stability, and faithfulness.”

Important: This is a research-positioning statement. Before presenting it as a novelty claim, perform a current literature search to verify the exact combination and any proposed stability formulation.

21. Architecture Summary

EEG Emotion → TIME + FREQUENCY + SPACE → Frequency-Aware Fusion → Transformer → Emotion Prediction → Temporal/Frequency/Channel XAI → Neurophysiological Interpretation → Stability + Faithfulness

Note: Hyperparameters, preprocessing choices, and explanation methods should be finalized after dataset-specific validation and a focused current literature/novelty review.